

CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy

Article type: Dataset and Software Article

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Author contribution statement

G.S. conceived the dataset and was its intellectual lead: he designed the label scheme, implemented the processing pipeline and the quality-control procedures, and wrote the manuscript. A.S. led the data annotation, annotated data, and edited the manuscript. A.T., M.K., R.C., D.H., F.M., H.S., M.S., S.S., M.M. and J.K. annotated data. M.I. and N.A. performed the Castellvi grading, each reading every case independently and resolving disagreements by consensus, adjudicated the candidate instrumentation, and trained the student annotators in CT reading. All authors read and approved the final version of the manuscript.

Conflict of interest statement

The authors have no relevant conflicts of interest to disclose.

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Data and code availability (unblinded)

Dataset, archived before submission: <https://doi.org/10.5281/zenodo.22139642>

(concept identifier, resolves to the current version); the version described in the manuscript is v8, <https://doi.org/10.5281/zenodo.22293878>.

Code: <https://github.com/Gregory-Schwing-MD-PhD/CTSpinoPelvic1K>, tagged at the release commit.

Mirror: <https://huggingface.co/datasets/OpenSpineConsortium/CTSpinoPelvic1K>, a convenience mirror, not the archive of record.

Pseudolabel model checkpoints (Sec. II.F of the manuscript):

<https://huggingface.co/OpenSpineConsortium/spinopelvic-seg-checkpoints>, with

the training and inference code at

<https://github.com/OpenSpineConsortium/spinopelvic-seg>.

Ethics

This work uses publicly available, de-identified imaging and annotations derived from it.

The Wayne State University Institutional Review Board issued a Notice of IRB

Administrative Determination of Non-Human Participant Research on 2 September 2026

(WSU IRB HPR Number 2026-269, “Open Spine Consortium: Secondary Analysis of

Publicly Available, De-Identified Imaging Datasets and the Derived Annotations,

Software, and Models Produced From Them”), finding that the project does not

constitute human participant research under the Common Rule at 45 CFR 46 and that

IRB review and oversight are not required.